Takehome Questions — Noni Ponyu

DS-413 — B22054

1) In the current era dominated by LLMs & deep learning what is the significance of fundamental machine learning models?

→ Fundamental machine learning models remain significant because they are

(i) Effecient and interpretable

(ii) Foundational for understanding

(iii) Effective for small Data

(iv) Real world applications.

2) Gradient descent questions in online form

→ $g(w) = aw^2 + b \quad w \in R \quad a = 1$

$$w_{t+1} = w_t - \eta \nabla g(w_t)$$

$$w_{t+1} = w_t - 2w_t = -w_t$$

3 iteration

(1) — $w_2 = w_1 = -5$

(2) — $w_3 = -w_2 = -(-5) = 5$

(3) — $w_4 = -w_3 = -5$

3) Normalization is very important why?

→ Because it involves scaling the data value in a certain range which avoids outliers.

4) Is lasso better or Ridge for subset selection

→ Lasso is better for subset selection as it shrinks coeffecients continuously without making them zero.

5) What does elastic net do?

→ Elastic net provides a middle ground containing L_1 & L_2

6) Why l_1-norm regularization gives sparse solution compared to l_2 norm

→ L_1 gives sparse solution because its penalty term, creates a sharp corner at origin. L_2 uses β_i^2 which forms a smooth, circular constraint.

7) $$\hat{\sigma}^2_{MLE} = \frac{1}{\frac{N-1}{N}} \times \hat{\sigma}^2_{MLE}$$

$$= \frac{1}{\frac{N-1}{N}} \times \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2_{ML}$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{N-1} \times \sum_{i=1}^{N} (x_i - \mu_{ML})^2$$

8) Can we use different PDFs for different class conditional PDFs?

⇒ Yes, as different class may have distinct data distribution

Ex: Gaussian, Poisson

9) ⇒ Dual is computationally efficient than primal

10) How to learn a tree with continuous input feature value?

→ To train a decision tree with continuous input features, split the features space using threshold that minimize impurity. This process ensure the tree can adapt to continuous inputs & make accurate predictions.

(11) Decision Tree

Gender	Major	Like
M	Math	Yes
F	History	No
M	CS	Yes
F	Math	No
F	Math	No
M	CS	Yes
M	History	No
F	Math	Yes

$$H(like) = -\frac{4}{8}\log\left(\frac{4}{8}\right) - \frac{4}{8}\log\left(\frac{4}{8}\right) = 1$$

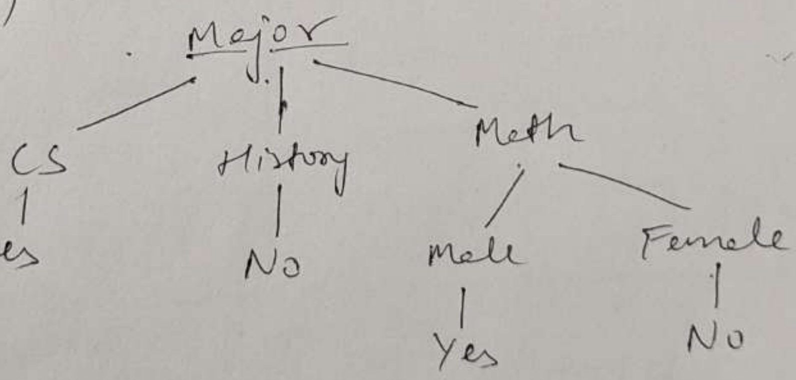
$$H(male) = -\frac{3}{4}\log\left(\frac{3}{4}\right) - \frac{1}{4}\log\left(\frac{1}{4}\right) = 0.81$$

$$H(female) = -\frac{1}{4}\log\left(\frac{1}{4}\right) - \frac{3}{4}\log\left(\frac{3}{4}\right) = 0.81$$

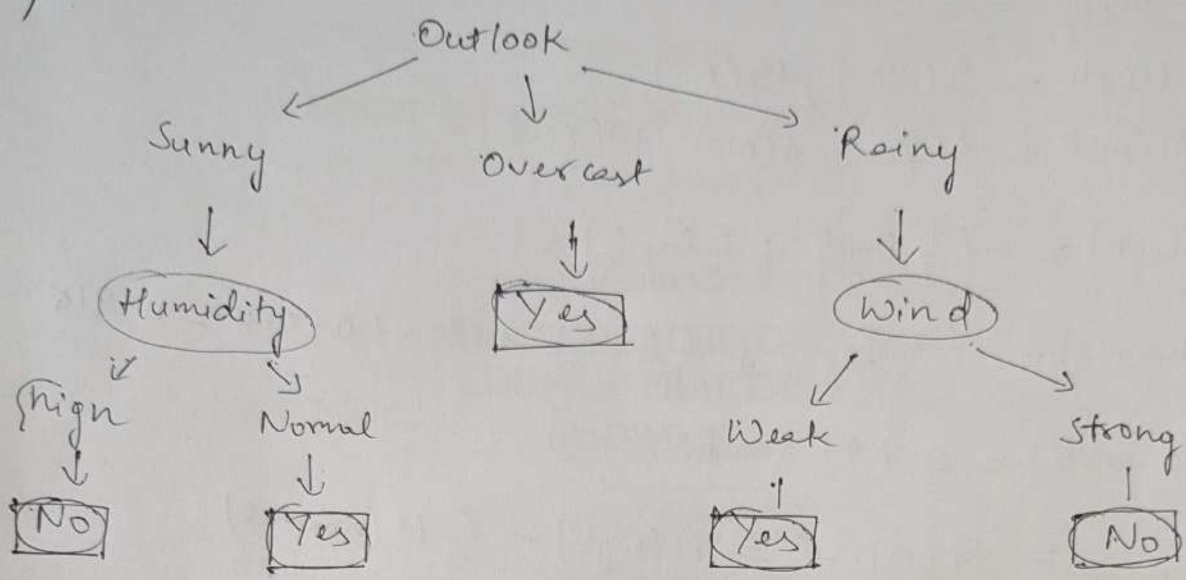
$H(Gender) = 0.81$, $\cancel{H(History) = 0.5}$

$H(math) = 1$, $H(CS) = 0$

$H(major) = 0.5$



12)



For outlook = rainy, data entries are

Outlook	Temperature	Humidity	Wind	Play Tennis
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Rainy	Mild	Normal	Weak	Yes
Rainy	Mild	High	Strong	No

For Temperature

Mild : 4(Yes), 10(Yes), 14(No)

Cool : 5(Yes), 6(No)

$$H(\text{mild}) = -\left(\frac{2}{3}\log\frac{2}{3} + \frac{1}{3}\log\frac{1}{3}\right) = 0.390 + 0.528 = 0.918$$

$$H(\text{cool}) = -\left(\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right) = 1$$

$$H(\text{Overall}) = -\left(\frac{3}{5}\log\frac{3}{5} + \frac{2}{5}\log\frac{2}{5}\right) = 0.442 + 0.529 = 0.971$$

$$IG_{temperature} = H(overall) - \frac{3}{5}H(mild) - \frac{2}{5}H(cool)$$
$$= 0.0202$$

For Humidity:

High = 4(Yes), 14(No)

Normal = 5(Yes), 6(No), 10(Yes)

$$H(high) = -\left(\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right) = 1$$

$$H(normal) = -\left(\frac{2}{3}\log\frac{2}{3} + \frac{1}{3}\log\frac{1}{3}\right) = 0.390 + 0.528 = 0.918$$

$H(overall) = 0.971$ (as previous)

$$IG_{humidity} = H(0) - \frac{2}{5}H(high) - \frac{3}{5}H(normal)$$
$$= 0.971 - \frac{2}{5}(1) - \frac{3}{5}(0.918)$$
$$= 0.0202$$

For wind:

Week: 4(Yes), 5(Yes), 10(Yes)

strong: 6(No), 14(Yes)

$H(week) = 0$

$H(strong) = 0$

$H(overall) = 0.971$

$$IG_{wind} = H(overall) - \frac{2}{5}H(strong) - \frac{3}{5}H(week)$$
$$= 0.971$$

IG_{wind} is maximum, we choose wind. Further both subset has $H=0$; i.e subsets are pure. Hence stop the tree here.